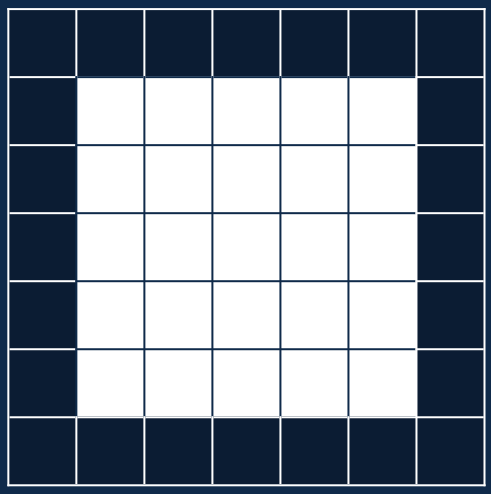
COMPUTER GRAPHICS · LAB TASK

TASK 01

شبكة مربعات ١ سم بإطار أسود

Draw a window filled with empty white squares of 1 cm, surrounded by a frame of black squares — OpenGL / GLUT.



Computer Graphics and Man-Machine Interface · Dr. Manar Donia

الكود
Code

Q1

Draw a window filled with empty white squares of 1 cm (38 pixel), surrounded by a frame of black squares.

```
#include<windows.h>
#include<GL/gl.h>
#include<GL/glut.h>
#include<iostream>
#include<math.h>
using namespace std;


void myInit();
void myDisplay();

int main(int argc, char** argv) {
    glutInit(&argc, argv);
    glutInitDisplayMode(GLUT_RGB);
    glutInitWindowPosition(100, 150);
    glutInitWindowSize(532, 532);
    glutCreateWindow("sasa_window");
    myInit();
    glutDisplayFunc(myDisplay);
    glutMainLoop();
    return 0;
}


void myInit() {
    glClearColor(1.0, 1.0, 1.0, 1.0);
    glPointSize(10.0);
    glMatrixMode(GL_PROJECTION);
    glLoadIdentity();
    gluOrtho2D(-266, 266, -266, 266);
}


void myDisplay() {
    glClear(GL_COLOR_BUFFER_BIT);

    glColor3f(0, 0, 0);
    glBegin(GL_POLYGON);
    glVertex2i(-266, -266);
    glVertex2i(266, -266);
    glVertex2i(266, 266);
    glVertex2i(-266, 266);
    glEnd();

    glColor3f(1.0, 1.0, 1.0);
    glBegin(GL_POLYGON);
    glVertex2i(-228, -228);
    glVertex2i(228, -228);
    glVertex2i(228, 228);
    glVertex2i(-228, 228);
```

```
    glEnd();

    glColor3f(1.0, 1.0, 1.0);
    glBegin(GL_LINES);
    for (int i = -228; i < 266; i = i + 38) {
        glVertex2i(i, -266);
        glVertex2i(i, -228);
        glVertex2i(i, 228);
        glVertex2i(i, 266);
        glVertex2i(-266, i);
        glVertex2i(-228, i);
        glVertex2i(228, i);
        glVertex2i(266, i);
    }
    glEnd();

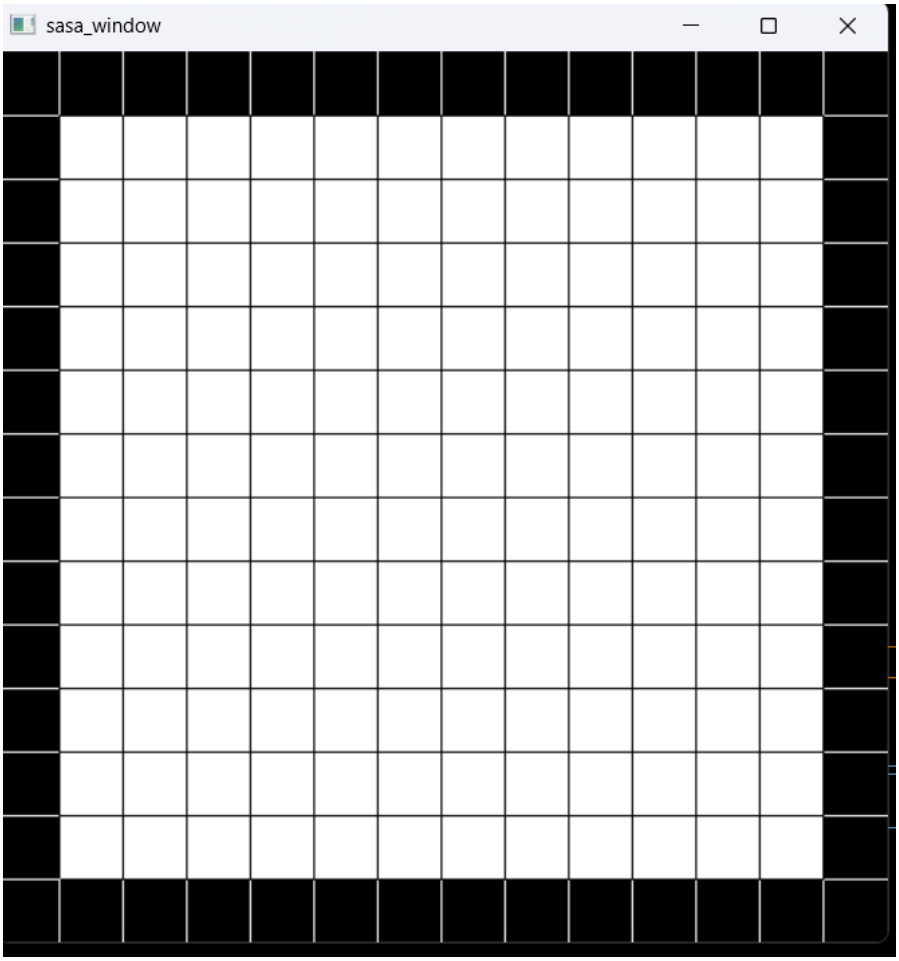
    glColor3f(0, 0, 0);
    glBegin(GL_LINES);
    for (int i = -228; i <= 228; i = i + 38) {
        glVertex2i(i, -228);
        glVertex2i(i, 228);
        glVertex2i(-228, i);
        glVertex2i(228, i);
    }
    glEnd();

    glFlush();
}
```

الناتج
Output

OUT

Output: the program window (sasa_window), 532 × 532 pixel = 14 × 14 squares.



الفكرة بنرسم مربع أسود قد الشاشة، وجواه مربع أبيض أصغر بمربع من كل ناحية. بعدين خطوط بيضا تقسّم الإطار لمربعات سودا، وخطوط سودا تقسّم الجزء الأبيض لمربعات بيضا.

1. **gluOrtho2D(-266, 266, -266, 266):** world window centred on (0,0); 1 unit = 1 pixel.
2. **GL_POLYGON:** the black frame, then the white area inside it (from −228 to 228).
3. **GL_LINES:** white lines divide the frame, black lines divide the inside, every 38 pixel.
4. **38 pixel ≈ 1 cm:** 96 pixel per inch ÷ 2.54 cm.